

# Tools and Technologies

The CryorApply application is built using a modern, reliable, and cost-effective technology stack. The system is split into three simple parts: the client layer (what the user sees), the backend layer (the server logic), and external managed services.

## Summary Stack Table

| Application Layer    | Technology                  | Purpose                                                                         |
|----------------------|-----------------------------|---------------------------------------------------------------------------------|
| Frontend (Interface) | Next.js 15, Tailwind CSS    | The visual look of the website, page routing, and responsive design.            |
| Backend (Server)     | Node.js, Next.js API routes | Main business logic, handling HTTP requests, and connecting system components.  |
| AI Service           | Google Gemini 3.1 Flash     | The core AI engine that analyzes the CV text and generates structured feedback. |
| Database             | Supabase (PostgreSQL)       | Relational storage for user accounts, upload records, and review results.       |
| File Storage         | Supabase Storage            | A secure cloud storage folder for keeping the uploaded CV files.                |
| Auth (Login)         | Supabase Auth               | Secure user registration and sign-in using token authentication.                |

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| Deployment | Vercel / Docker | Hosting the website online and ensuring a stable running environment. |
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## Detailed Component Breakdown

### 1. Interface (Frontend)

- **Next.js 15:** This is the main framework for the visible part of the website that users interact with. It handles routing and helps users easily navigate between three main screens: the sign-in page, the CV drag-and-drop upload page, and the results page.
- **Tailwind CSS:** A tool used to style the interface quickly. It makes the website look modern and work smoothly on both mobile phones and desktop computers, controlling the layout of feedback cards and buttons.

### 2. Server (Backend & API Gateway)

- **Node.js & Next.js API Routes:** This is the hidden "brain" of the app running on the server. It acts as an API gateway that receives requests, processes the uploaded files, extracts raw text content from PDF or DOCX formats, and prepares it for the AI.
- **JWT Middleware:** A digital security guard on the server. It checks the user's login tokens to make sure no unauthorized person can upload files or view someone else's personal review history.

### 3. Artificial Intelligence (AI Engine)

- **Google Gemini 3.1 Flash API:** The main AI engine that evaluates the resume. It was chosen because it is very fast, highly cost-effective, and can process large documents easily. It reads the CV text, scores it from 0 to 100, and organizes feedback into four categories: structure, skills, experience, and formatting. It sends the data back in a clean JSON format so the frontend can display it beautifully.

### 4. Data, Security, and Infrastructure

- **Supabase (PostgreSQL):** Our relational database. It is a reliable digital warehouse where user profiles, file paths, and detailed AI feedback tables (using JSONB formats) are safely stored.
- **Supabase Storage:** A secure cloud storage bucket where the original CV files (up to 5 MB) are uploaded. These files are protected and linked directly to each specific user's account.
- **Vercel:** The cloud platform where our website runs on the internet. It connects directly to our GitHub repository and automatically updates the website whenever the team updates the code.
- **Docker:** A tool that packages the whole project into an independent virtual container. This ensures the program runs exactly the same way on every developer's local computer and on the final production server.

